

MetCouncil Building Sector Interventions: Explanation of Equations

Part I: Overall projection of total GHG from the building sector by 2040 (Cell D42 in spreadsheet)

- **Short description:** This is the overall total community-wide GHG counted for 2040, assuming the grid is still like the 2018 grid, because we show grid decarbonization as an intervention. The formula is basically a) electricity emission factor multiplied by residential, commercial, and industrial electricity use intensities in units of square footage or per employee as appropriate, multiplied by the total square footage or the total employees as appropriate; b) the same calculation as for electricity, but for natural gas.
- **Formula:**
$$= D39 * (D7 * D15 * D18 / 1000 + D25 * D29 + D31 * D34) + D40 * (D7 * D15 * D19 + D25 * D30 + D31 * D35)$$
- **Formula Term 1** = D39 = Electricity emission factor in 2018; kg CO₂e/MWh; used also in 2040 calculations before grid decarbonization
- **Formula Term 2** = $(D7 * D15 * D18 / 1000 + D25 * D29 + D31 * D34)$ = Total population in 2040 * Residential floor area per capita, sqft/person, in 2040 * Residential electricity per floor area, kWh/sqft / 1000 + Number of commercial workers in 2040 (from Minnesota Quarterly Census of Employment and Wages) * Commercial electricity use per worker in 2040, MWh/worker/yr + Number of industrial workers in 2040 * Industrial electricity use per worker in 2040, MWh/worker = total electricity use, community-wide
- **Formula Term 3** = Gas emission factor in 2040; units are kg CO₂e/Therm
- **Formula Term 4** = $(D7 * D15 * D19 + D25 * D30 + D31 * D35)$ = Total population in 2040 * Residential floor area per capita, sqft/person, in 2040 * Residential gas per floor area, therm/sqft + Number of commercial workers in 2040 (from Minnesota Quarterly Census of Employment and Wages) * Commercial gas use per worker in 2040, therm/worker/yr + Number of industrial workers in 2040 * Industrial gas use per worker in 2040, therm/worker = total gas use, community-wide

Sources of Emission Factors

- Electricity emission factor (2018): 566.4 kg CO₂e/MWh - This emission factor is from eGrid data for the MRO west region: <https://www.epa.gov/egrid/download-data>.
 - The baseline keeps the same emission factor in 2040. Grid decarbonization by 80% is represented in the grid decarbonization wedge (see interventions).
- Gas emission factor (2018, remains unchanged for 2040): 5.31 kg CO₂e/Therm - From EPA GHG emission factor table. https://www.epa.gov/sites/default/files/2021-04/documents/emission-factors_apr2021.pdf

Part II: Interventions

1. MetCouncil Plan multi-family housing effects

- **Short description:** This equation calculates the impact of the MetCouncil's plan on total residential square footage, assuming no change in electricity grid (which is shown as a separate wedge). *This wedge can be deleted entirely—the only reason it is here is to show the impact of the existing plans adopted by the MetCouncil region.* Note: All the percentage reductions are calculated based on estimates of the MetCouncil's BAU projections, but how this differs from historical trends may be interesting to show.

MetCouncil's plan is increasing the percentage of multi-family housing substantially by 2040, which in turn tends to be smaller (1786 square feet per home (townhomes), compared to 2387 square feet for single-family homes, as obtained from tax parcel data), so the combination of the two has a large impact on housing floor area. Our trend analysis shows that the multi-family square footage has historically remained about constant, so our 2040 square footage for multi-family housing remains the same as 2018, at 1786 square feet. However, the historical trend from 2010 to 2018 showed single-family square footage increasing sharply, at an estimated rate of 1.1% per year (simple average over 10 years) (see the "Bldgs trent10to18 tab" in spreadsheet). We applied the same rate but capped it at a total 15% increase to get the 2040 single-family floor area to be 2387 square feet, as computed in the spreadsheet (see column D).

The first part of the equation calculates the total floor area by modeling what would have happened if the percentage of multi-family and single-family housing would stay constant at 2018 levels (for Lake Elmo, 6% and 94% respectively), but with the multi-family and single-family floor areas of 2040.

- **Formula** = $(D8 * C14 * D13 + D8 * (1 - C14) * D10) * (C18 * C39 / 1000 + C19 * C40) - (D10 * D9 + D13 * D12) * (C18 * C39 / 1000 + C19 * C40)$
- **Formula Term 1** = $(D8 * C14 * D13 + D8 * (1 - C14) * D10)$ = total number of households in 2040 * % of multi-family homes in 2018 * average floor area of multi-family home in 2040 + total number of households * (% single-family homes in 2018) * average single-family home floor area in 2040 = total square footage in 2040 (units of square feet)
- **Formula Term 2** = $(C18 * C39 / 1000 + C19 * C40)$ = residential electricity use intensity per floor area * emission factor of the electricity grid in 2040/1000 (to get units of kilogram CO2e per square foot) + gas use intensity * gas emission factor (to get units of kilogram CO2e per square foot)
- **Formula Term 3** = $(D10 * D9 + D13 * D12)$ = Same as Formula Term 1, except that now it applies the MetCouncil plan proportion of multi-family and single-family homes
- **Formula Term 4** = $(C18 * C39 / 1000 + C19 * C40)$ = Same as Formula Term 2, because in this equation, we're only showing the floor area effects, assuming the energy mix in 2040 is the same as 2018. The change in the grid is shown as a separate wedge.

2. Compact buildings (50% new construction is multi-family (MF))

- **Short description:** This strategy goes beyond the MetCouncil's 2040 plan, modeled in the previous strategy, by assuming that 50% of all new single-family construction will now become

multi-family, with the multi-family homes having smaller square footage, as explained previously.

- **Formula** = $(D9-C9) * M8 * (D10-D13) * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $D9-C9$ = number of single-family housing units in 2040 – number of single-family housing units in 2018 = new single-family housing construction between 2018 and 2040 (lowest estimate)
- **Formula Term 2** = $M8$ = percentage of new single-family converted to multi-family. This can be specified by the user (we have defaulted to 50%, as an aggressive scenario).
- **Formula Term 3** = $D10-D13$ = average floor area of single-family homes in 2040 – average floor area of multi-family homes in 2040 = reduction in floor area due to conversion for an average single housing unit from single-family to multi-family
- **Formula Term 4** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

3. Tree canopy energy reduction

- Tree canopy reductions were too small to model in a spreadsheet, due to their complexity.

4. Affordable floor area

- **Short description:** The original intent was to reduce the growth in floor area, recognizing that the cost of electricity will more than double in a net-zero carbon grid (according to Princeton University's [Net-Zero America report](#)), and raise the idea of an affordable floor area. Typically, energy burden is expressed when greater than 6% of income is used for energy services, but this number is difficult to evaluate, because we do not have income distribution data in a 2040 future city. Therefore, we instead modeled that about 50% of the single-family homes will respond to increased energy prices by decreasing their living space. This is a model assumption, which cities can adjust.
- **Formula** = $(D9-C9) * 0.5 * (D10 - (1 + M10) * C10) * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $D9-C9$ = number of single-family housing units in 2040 – number of single-family housing units in 2018 = new single-family housing construction between 2018 and 2040 (lowest estimate)
- **Formula Term 2** = 0.5 = The model assumes half the population will respond to doubling of energy prices by reducing living space of new single-family homes.
- **Formula Term 3** = $(D10 - (1 + M10) * C10)$ = 2040 average single-family home floor area – (1 + growth of single-family floor area [range between 0-5]) * average single-family floor area per household in 2018 = reduction in floor area of single-family homes due to presumed sensitivity to doubling the cost of energy services in a net-zero grid, as shown in Princeton University's [Net-Zero America report](#).
- **Formula Term 4** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

5. New homes (50%) are LEED-Gold

- **Short description:** Here we calculated the effect of 50% of new homes being constructed to LEED Gold standards.
- **Formula** = $M11 * (D9 - C9) * D11 * 0.64 * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $M11$ = % of new homes that are LEED Gold – user can specify a range of 0-100; model is defaulted to 50%.
- **Formula Term 2** = $D9 - C9$ = number of single-family housing units in 2040 – number of single-family housing units in 2018 = new single-family housing construction between 2018 and 2040 (lowest estimate)
- **Formula Term 3** = $D11 * 0.64$ = average floor area of new single-family homes * 64% reduction in energy use intensity due to highly-efficient LEED Gold construction
- **Formula Term 4** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

6. Existing homes (80%) retrofitted

- **Short description:** The previous strategies focused on new construction. This strategy focuses on existing homes and assumes 80% of them are retrofitted to reduce energy use by 33%.
- **Formula** = $(C9 * M12 * 0.33 * C10 + C12 * M12 * 0.33 * C13) * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $C9 * M12 * 0.33 * C10$ = number of single-family housing units in 2018 * percent of existing homes to be retrofitted (can be adjusted by user – defaulted here to 80% to represent action over 22 years) * 33% reduction in energy use intensity, due to efficiency gains of the retrofitting * 2018 average floor area of single-family homes
- **Formula Term 2** = $C12 * M12 * 0.33 * C13$ = number of multi-family homes in 2018 * percent of existing homes to be retrofitted * 33% reduction in energy use intensity * average floor area of multi-family home in 2018. Formula Term 1 + Formula Term 2 = total square footage of existing single- and multi-family homes that are under normal retrofit
- **Formula Term 3** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

7. Existing homes (20%), passive homes

- **Short description:** Same as previous strategy, except now the remaining homes are retrofitted to high passive house standards, which results in 66% reduction in energy use intensity (ZeroEnergy Design, 2021). For remaining details, see above.
- **Formula** = $M13 * (C9 * C10 + C12 * C13) * 0.66 * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $M13$ = % of largest existing homes that are retrofitted to passive home standards, assumed here to be 20%
- **Formula Term 2** = $(C9 * C10 + C12 * C13)$ = (number of single-family housing units in 2018 * 2018 average floor area of single-family homes) + (number of multi-family homes in 2018 * average floor area of multi-family home in 2018) = total floor area
- **Formula Term 3** = 0.66 = 66% reduction in energy use intensity due to passive house standards

- **Formula Term 4** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

8. Existing commercial buildings (80%), LEED-Gold

- **Short description:** This models the reduction in energy use intensity (EUI) that will result if 80% of existing commercial buildings are certified as LEED Gold, which is assumed to reduce EUI by 25%.
- **Formula** = $M14 * C25 * 0.25 * (C29 * D39 + D30 * D40)$
- **Formula Term 1** = $M14 * C25 * 0.25$ = % of existing commercial buildings are LEED Gold (user-specified, defaulted here to 80%) * 2018 number of commercial workers (from Minnesota Quarterly Census of Employment and Wages) * 25%, the assumed reduction in energy intensity
- **Formula Term 2** = $(C29 * D39 + D30 * D40)$ = [2018 commercial electricity use per worker, MWh/worker/year * electricity emission factor in 2040; kg CO2e/MWh] + [2040 commercial gas use per worker, therm/worker/year * gas emission factor in 2040; kg CO2e/Therm]

9. Behavioral & smart nudges (100% HH)

- **Short description:** This strategy assumes that all homes receive effective messages which encourage them to reduce energy use via nudges such as in-home display and smart meters. Such nudges can reduce household energy use by 11%, based on Gilbert et al. (ACEEE 2010).
- **Formula** = $M15 * D7 * D15 * 11\% * (D18 * D39 / 1000 + D19 * D40)$
- **Formula Term 1** = $M15 * D7 * D15 * 11\%$ = % of homes that change behaviors, assumed to be 100% in this model * total population in 2040 * Residential floor area per capita, sqft/person in 2040 * 11%, the assumed percentage reduction in household energy use as a result of smart nudges
- **Formula Term 2** = $(D18 * D39 / 1000 + D19 * D40)$ = emission factor for electricity and gas use, in units of kilograms of Co2e per square foot, described previously in the first strategy, and derived from energy use intensity data

10. Smart grid efficiency (100% of commercial buildings)

- **Short description:** Achieving a net-zero grid require transitioning to a smart grid. While the smart grid often helps in managing the load, studies have also shown that it can reduce energy use by approximately 11%, based on PNNL's "The Smart Grid: An Estimation of the Energy and CO2 Benefits." This 11% will apply across the board to anyone connected to the electricity grid. In this strategy and the next, we applied this percentage reduction to 100% of the electricity use. This strategy applies to commercial buildings because the residential reductions are already incorporated in the previous strategy. Note: All commercial and industrial energy use is computed on the basis of employees, not square footage.
- **Formula** = $M16 * (D25 * D29 - 0.8 * C25 * 0.25 * C29) * 11\% * D39$
- **Formula Term 1** = $M16$ = % of commercial buildings in the smart grid, assumed to be 100% (should not be changed by users of this model)
- **Formula Term 2** = $(D25 * D29 - 0.8 * C25 * 0.25 * C29)$ = [Number of commercial workers in 2040 (from Minnesota Quarterly Census of Employment and Wages) * Commercial electricity use per worker, MWh/worker/yr for 2040] – [0.8 * Number of commercial workers in 2018 * 25%

reduction in electricity use because of LEED Gold buildings [see previous strategy, where 80% of commercial buildings adopt LEED standards] * Commercial electricity use per worker, MWh/worker/yr for 2018] = effectively calculating remaining electricity after 80% of commercial buildings adopt LEED Gold, which is assumed to reduce energy use by 25%

- **Formula Term 3** = $11\% * D39$ = reduction in energy use due to smart grid (see above) * electricity emission factor in 2040 in kilograms of CO₂e/MWh

11. Smart grid efficiency (100% of industries)

- **Short description:** Smart grid and technologies are projected to reduce industrial electricity use by 11% .
- **Formula** = $M17 * D31 * D34 * 11\% * D39$
- **Formula Term 1** = $M17 * D31 * D34$ = % of industries in the smart grid, assumed to be 100% (should not be changed by users of this model) * Number of industrial workers in 2040, based on trends from 2010 to 2018 (due to the rapid development of this city, the rate is reduced to half that of the period between 2010 and 2018) * projected industrial electricity use per worker, MWh/worker, for 2040
- **Formula Term 2** = $11\% * D39$ = % reduction in energy use due to smart grid (see above) * electricity emission factor in 2040 in kilograms of CO₂e/MWh

12. Grid decarbonization, 113 kgCO₂e/MW

- **Short description:** This wedge demonstrates the impact of grid decarbonization after all the efficiency reductions are subtracted to yield the final electricity demand estimate. Note: Switching to electricity is shown as a separate scenario after grid decarbonization.
- **Formula** = $((D7 * D15 * D18 + D25 * D29 * 1000 + D31 * D34 * 1000) - \text{SUM}(A8:A17)) * (D39 / 1000) * M18$
- **Formula Term 1** = $D7 * D15 * D18$ = Total population in 2040 * Residential floor area per capita in 2040, sq ft/person * Residential electricity per floor area in 2040, kWh/sqft = residential electricity used in 2040, prior to efficiency improvements but capturing historic trends in EUI reduction. Units are in kilowatt hours.
- **Formula Term 2** = $D25 * D29 * 1000$ = Number of commercial workers in 2040 (from Minnesota Quarterly Census of Employment and Wages) * Commercial electricity use per worker in 2040, MWh/worker/yr * 1000 = Commercial electricity used in 2040, prior to efficiency improvements but capturing historic trends in EUI reduction. We multiply by 1000 to retain units in kilowatt hours.
- **Formula Term 3** = $D31 * D34 * 1000$ = Number of industrial workers in 2040 * Industrial electricity use per worker in 2040, MWh/worker, * 1000 = Industrial electricity used in 2040, prior to efficiency improvements but capturing historic trends in EUI reduction. We multiply by 1000 to retain units in kilowatt hours.
- **Formula Term 4** = $\text{SUM}(A8:A17)$ = sum of all previously described interventions in kilowatt hours, as shown in column AE (because you cannot decarbonize energy that is not used due to these previous interventions)
- **Formula Term 5** = $(D39 / 1000) * M18$ = Electricity emission factor in 2040 in kg CO₂e/MWh * 1/1000 to convert to kg CO₂e/kWh * % reduction in grid emission factor from 2018 (assumed to be 80% due to grid decarbonization), according to Xcel Energy goals

13. Electrifying heating (59% of gas-heating households transition to electric heating)

- Short description:** This equation calculates reduction in energy use and GHG emissions due to electrifying heating in households (ex. switching to ground source heat pumps and tankless water heaters), assuming that by 2040, 59% of households will use electric heating, based on the [Net-Zero America report](#). Currently, 14% of households already use electricity for heating, according to American Community Survey estimates (in Washington county). Therefore, a transition of 45% of homes to electric heating (59%-14%) will have the modeled effects on GHG emissions. Note: In regions of Minnesota, 71% of annual household gas is used for space heating, and 24% of gas is used for water heating, according to the [Residential Energy Consumption Survey \(RECS\) \(2015\)](#):
 - Formula Term 1:** Models space heating transitions to heat pumps and the avoided emissions from gas use, after subtracting reductions in gas use due to efficiency strategies described previously (summed up in cell AF23)
 - Formula Term 2:** Describes increase in electricity use in space heating due to switching to electrification and associated emissions, applying some conversion factors for the assumed efficiency of natural gas boilers today, and the efficiency of the heat pumps that will replace them. Gas boiler efficiency is assumed to be 80%. Ground source heat pump COP is 5.2. Because the grid still has 20% GHG emissions relative to 2018, the switch to electricity will not zero out the gas emissions.
 - Formula Terms 3 and 4:** Same as above, except they are applied for water heating, where RECS data show that 24% of annual gas use is for water heating, of which (59%-14%) will be switched to electric. Electric tankless heater energy efficiency is 1.
- Formula** = $(D7 \cdot D15 \cdot D19) \cdot M19 \cdot D40 \cdot (1 - AF23) \cdot 71\% - (D7 \cdot M19 \cdot D15 \cdot D19 \cdot 71\% \cdot (1 - AF23) \cdot 0.8/5.02) \cdot 0.0293 \cdot D39 \cdot 0.2 + (D19 \cdot D7 \cdot D15 \cdot 24\% \cdot M19 \cdot D40 \cdot (1 - AF23) - D19 \cdot D7 \cdot D15 \cdot 24\% \cdot M19 \cdot (1 - AF23)) / 1 \cdot D39 \cdot 0.0293 \cdot 0.2$
- Formula Term 1** = $(D7 \cdot D15 \cdot D19) \cdot M19 \cdot D40 \cdot (1 - AF23) \cdot 71\%$ = [Total population in 2040 * Residential floor area per capita in 2040, sqft/person * Residential gas per floor area in 2040, therm/sqft] * % of additional households with heating electrified (assumed to be (59%-14%)) * Gas EF in 2040; kg CO2e/Therm * [1-% gas saved in residential energy use]*71% of homes using gas for space heating, based on RECS data
- Formula Term 2** = $(D7 \cdot M19 \cdot D15 \cdot D19 \cdot 71\% \cdot (1 - AF23) \cdot 0.8/5.02) \cdot 0.0293 \cdot D39 \cdot 0.2$ = Total population in 2040 * % of additional households with heating electrified (calculated as 59%-14%) * Residential floor area per capita in 2040, sq ft/person * Residential gas per floor area in 2040, therm/sq ft * 71% (fraction of total gas use for space heating) * [1-% gas saved in residential energy use] * (0.8/5.02) (the ratio of boiler efficiency/heat pump efficiency) * 0.0293, representing the conversation factor from therm to MWh, with 1 therm = 0.0293 MWh.
- Formula Term 3** = $(D19 \cdot D7 \cdot D15 \cdot 24\% \cdot M19 \cdot D40 \cdot (1 - AF23))$ - Residential gas per floor area, therm/sq ft * Total population in 2040 * Residential floor area per capita in 2040, sq ft/person * 24% of annual gas for water heating * % of additional households with heating electrified (calculated as 59% - 14%) * Gas emission factor in 2040; kg CO2e/Therm * reduction in gas use due to efficiency measures

- **Formula Term 4** = $D19 * D7 * D15 * 24\% * M19 * (1 - AF23)$ = Residential gas per floor area in 2040, therm/sq ft * Total population in 2040 * Residential floor area per capita in 2040, sq ft/person * 24% of annual gas used for water heating * % of additional households with heating electrified (assumed to be 59%-14%) * reduction in gas use due to efficiency measures /1 (energy efficiency of electric water heater) * 0.0293 (converting therm to MWh) * (D39*0.2), the emission factor of the decarbonized grid in 2040

14. Renewable natural gas (RNG) substitutes 31.7% of natural gas in all sectors

- **Short description:** The remaining homes that did not transition to electric space and water heating are assumed to have switched to renewable natural gas, based on the State of Minnesota's policy to aggregate all waste-to-energy project towards renewable natural gas. This includes municipal solid waste, agricultural waste, etc., converted to biogas, which is assumed to be carbon neutral.
 - We assume that RNG will substitute 31.7% of all natural gas remaining in all the sectors. According to the American Biogas Council's estimate, Minnesota can produce 46.91 billion cu.ft renewable natural gas, which is 47 PJ. According to the Net-Zero America Report, 148 PJ will be needed in 2040 in the high efficiency and renewable scenario. Therefore, we assume that RNG can replace 31.7% of remaining natural gas use at the state level. Estimates for New York State are available in Taboada et. al, 2021 (see references).
- **Formula** = $(D7 * D15 * D19 + D25 * D30 + D31 * D35 - SUM(AF8:AF19)) * M20 * D40$
- **Formula Term 1** = $D7 * D15 * D19 + D25 * D30 + D31 * D35$ = [Total population in 2040 * Residential floor area per capita in 2040, sq ft/person * Residential gas per floor area in 2040, therm/sq ft] + [Number of commercial workers in 2040 (from Minnesota Quarterly Census of Employment and Wages) * Commercial gas use per worker in 2040, therm/worker/yr] + [Number of industrial workers in 2040 * Industrial gas use per worker in 2040, therm/worker]
- **Formula Term 2** = $SUM(AF8:AF19)$ = sum of all previously described interventions in therm, as seen in column AF
- **Formula Term 3** = $M20 * D40$ = % of natural gas replaced by RNG (utility level), assumed to be 31.7%, as determined by MN potential * Gas emission factor in 2040; kg CO2e/Therm

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